Speech Processing -Worksheet 02 Solution

Exercise 1: System Properties Analysis

Given system: $y[n] = x[n] + 2$

1. Linearity Check

A system is linear if it satisfies superposition:

$$\mathcal{T}\{ax_1[n] + bx_2[n]\} = a\mathcal{T}\{x_1[n]\} + b\mathcal{T}\{x_2[n]\}$$

For the given system:

$$\mathcal{T}\{x[n]\} = x[n] + 2$$

Let

$$\mathcal{T}\{x_1[n]\} = x_1[n] + 2, \quad \mathcal{T}\{x_2[n]\} = x_2[n] + 2$$

Then,

$$\mathcal{T}\{ax_1[n] + bx_2[n]\} = ax_1[n] + bx_2[n] + 2$$

But

$$a\mathcal{T}\{x_1[n]\} + b\mathcal{T}\{x_2[n]\} = a(x_1[n] + 2) + b(x_2[n] + 2) = ax_1[n] + bx_2[n] + 2a + 2b$$

Since

$$ax_1[n] + bx_2[n] + 2 \neq ax_1[n] + bx_2[n] + 2a + 2b$$

$$\boxed{\textbf{The system is not linear.}}$$

2. Time-Invariance Check

Shift the input by n_0:

$$\mathcal{T}\{x[n - n_0]\} = x[n - n_0] + 2$$

Shift the original output:

$$y[n - n_0] = x[n - n_0] + 2$$

Since

$$\mathcal{T}\{x[n - n_0]\} = y[n - n_0]$$

$$\boxed{\textbf{The system is time-invariant.}}$$

Conclusion

$$\boxed{\text{The system is time-invariant but not linear.}}$$

Exercise 2: LTI System Analysis

Given: LTI system with $x[n] = u[n]$, $y[n] = 2^{-n}u[n]$

Step 1: Convolution Property

For an LTI system:

$$y[n] = x[n] * h[n]$$

Since $x[n] = u[n]$, we have

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, u[n-k]$$

Because $u[n-k] = 1$ for $k \leq n$ and 0 otherwise,

$$y[n] = \sum_{k=-\infty}^{n} h[k]$$

Thus, the output is the cumulative sum of $h[n]$, which implies

$$h[n] = y[n] - y[n-1]$$

Step 2: Compute $h[n]$

Given

$$y[n] = 2^{-n}u[n]$$

we compute

$$y[n-1] = 2^{-(n-1)}u[n-1]$$

Therefore,

$$h[n] = y[n] - y[n-1] = 2^{-n}u[n] - 2^{-(n-1)}u[n-1]$$

Since $2^{-(n-1)} = 2 \cdot 2^{-n}$,

$$h[n] = 2^{-n}u[n] - 2 \cdot 2^{-n}u[n-1]$$

Factor 2^{-n}:

$$h[n] = 2^{-n}\big(u[n] - 2u[n-1]\big)$$

Using $u[n] = \delta[n] + u[n-1]$, we obtain

$$\boxed{h[n] = \delta[n] - 2^{-n}u[n-1]}$$

Step 3: Stability Check

An LTI system is stable if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Here, $\delta[n]$ is finite and

$$\sum_{n=1}^{\infty} 2^{-n}$$

is a convergent geometric series.

$$\boxed{\textbf{The system is stable.}}$$

Exercise 3: Convolution of Rectangular Pulses

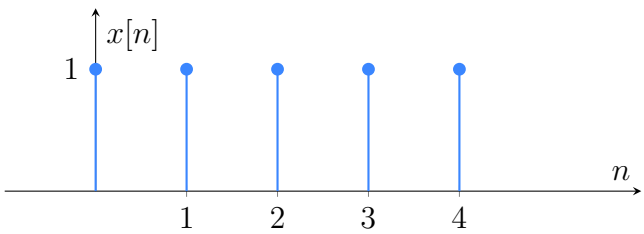
Given: $x[n] = u[n] - u[n-5]$

Signal Description

$x[n]$ is a rectangular pulse of length 5:

$$x[n] = \begin{cases} 1, & 0 \le n \le 4, \\ 0, & \text{otherwise} \end{cases}$$

Original Signal $x[n]$



Convolution Setup

We compute

$$y[n] = x[n] * x[n] = \sum_{k=-\infty}^{\infty} x[k]\, x[n-k]$$

Support of the Summation

Since $x[k]$ is non-zero only for $k \in [0, 4]$, and $x[n-k]$ is non-zero when

$$0 \le n - k \le 4 \quad \Longrightarrow \quad k \in [n-4,\, n]$$

the product $x[k]x[n-k]$ is non-zero only when the intervals overlap:

$$k \in [\max(0, n-4),\, \min(4, n)]$$

The result is non-zero only when these intervals intersect, which occurs for

$$n = 0, 1, \ldots, 8$$

Computation for Each n

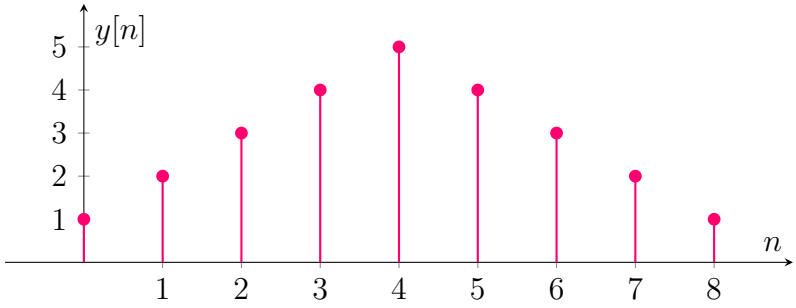
$$\begin{aligned}
n = 0: &\quad k = 0 \Rightarrow y[0] = 1, \\
n = 1: &\quad k = 0, 1 \Rightarrow y[1] = 2, \\
n = 2: &\quad k = 0, 1, 2 \Rightarrow y[2] = 3, \\
n = 3: &\quad k = 0, 1, 2, 3 \Rightarrow y[3] = 4, \\
n = 4: &\quad k = 0, 1, 2, 3, 4 \Rightarrow y[4] = 5, \\
n = 5: &\quad k = 1, 2, 3, 4 \Rightarrow y[5] = 4, \\
n = 6: &\quad k = 2, 3, 4 \Rightarrow y[6] = 3, \\
n = 7: &\quad k = 3, 4 \Rightarrow y[7] = 2, \\
n = 8: &\quad k = 4 \Rightarrow y[8] = 1
\end{aligned}$$

Final Result

$$y[n] = \begin{cases} n+1, & 0 \le n \le 4, \\ 9-n, & 5 \le n \le 8, \\ 0, & \text{otherwise} \end{cases}$$

$$\boxed{y[n] = \{1, 2, 3, 4, 5, 4, 3, 2, 1\} \text{ for } n = 0, \ldots, 8}$$

Convolution Result $y[n] = x[n] * x[n]$



Exercise 4: Convolution with Exponential and Rectangular Pulse

Given: $x[n] = e^{-|n|}$, $h[n] = u[n+5] - u[n-6]$

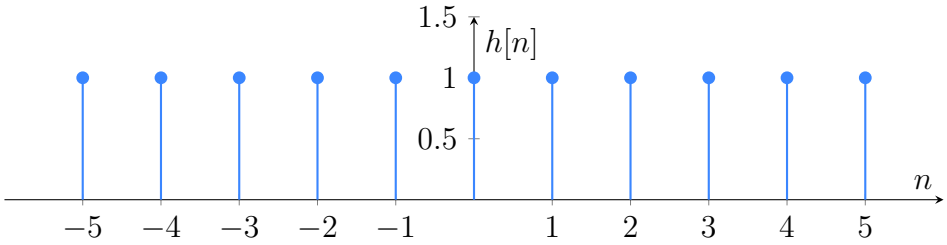
Step 1: Description of $h[n]$

Since

$$u[n+5] - u[n-6] = \begin{cases} 1, & -5 \le n \le 5, \\ 0, & \text{otherwise} \end{cases}$$

$h[n]$ is a rectangular pulse from $n = -5$ to $n = 5$ (length 11).

Impulse Response $h[n]$



Step 2: Convolution Formula

The convolution is

$$y[n] = \sum_{k=-\infty}^{\infty} e^{-|k|}\, h[n-k]$$

Since $h[n-k] = 1$ when

$$-5 \le n-k \le 5 \quad \Longleftrightarrow \quad n-5 \le k \le n+5$$

we obtain

$$y[n] = \sum_{k=n-5}^{n+5} e^{-|k|}$$

Case 1: $n \ge 5$

If $n \ge 5$, then $n-5 \ge 0$, so all k in the interval are non-negative and $e^{-|k|} = e^{-k}$.

Thus

$$y[n] = \sum_{k=n-5}^{n+5} e^{-k}$$

Using the geometric series formula,

$$\sum_{k=p}^{q} e^{-k} = e^{-p}\frac{1-e^{-(q-p+1)}}{1-e^{-1}}$$

we obtain

$$\boxed{y[n] = e^{-(n-5)}\frac{1-e^{-11}}{1-e^{-1}}, \quad n \ge 5}$$

Case 2: $n \le -5$

If $n \le -5$, then $n+5 \le 0$, so all k are non-positive. Let $k = -m$, $m \ge 0$. Then $e^{-|k|} = e^{-m}$.

The limits become $m = -(n+5)$ to $-(n-5)$. Thus

$$y[n] = \sum_{m=-(n+5)}^{-(n-5)} e^{-m} = e^{n+5}\frac{1-e^{-11}}{1-e^{-1}}$$

$$\boxed{y[n] = e^{n+5}\frac{1-e^{-11}}{1-e^{-1}}, \quad n \le -5}$$

Case 3: $-5 \le n \le 5$

In this case, the interval $[n-5, n+5]$ contains both negative and positive indices. Split the sum at $k = 0$:

$$y[n] = \sum_{k=n-5}^{-1} e^{-|k|} + e^{0} + \sum_{k=1}^{n+5} e^{-|k|}$$

Negative part: Let $m = -k$. Then m runs from 1 to $5 - n$:

$$\sum_{k=n-5}^{-1} e^{-|k|} = \sum_{m=1}^{5-n} e^{-m} = \frac{e^{-1}\left(1 - e^{-(5-n)}\right)}{1 - e^{-1}}$$

Positive part:

$$\sum_{k=1}^{n+5} e^{-k} = \frac{e^{-1}\left(1 - e^{-(n+5)}\right)}{1 - e^{-1}}$$

Adding the term at $k = 0$:

$$\boxed{y[n] = 1 + \frac{e^{-1}}{1 - e^{-1}}\left(2 - e^{-(5-n)} - e^{-(n+5)}\right), \quad -5 \leq n \leq 5}$$

Exercise 5: System Connections

Given: $h_1[n] = e^{-an}u[n]$, $h_2[n] = e^{-bn}u[n]$, $h_3[n] = u[n]$

(a) Parallel Connection

For systems connected in parallel, the overall impulse response is the sum:

$$h[n] = h_1[n] + h_2[n] + h_3[n]$$

$$\boxed{h[n] = e^{-an}u[n] + e^{-bn}u[n] + u[n]}$$

(b) $h_1[n]$ in parallel with cascade of $h_2[n]$ and $h_3[n]$

Step 1: Cascade of $h_2[n]$ and $h_3[n]$

The cascade of two LTI systems is obtained by convolution:

$$h_{23}[n] = h_2[n] * h_3[n] = \sum_{k=-\infty}^{\infty} h_2[k]u[n-k]$$

Since $h_2[k] = 0$ for $k < 0$ and $u[n-k] = 0$ for $k > n$, for $n \geq 0$:

$$h_{23}[n] = \sum_{k=0}^{n} e^{-bk} = \frac{1 - e^{-b(n+1)}}{1 - e^{-b}}u[n]$$

Step 2: Add $h_1[n]$ in parallel

$$\boxed{h[n] = e^{-an}u[n] + \frac{1 - e^{-b(n+1)}}{1 - e^{-b}}u[n]}$$

Exercise 6: Deconvolution Problem

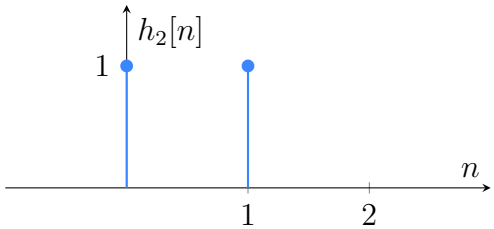
Given: $h[n] = h_1[n] * h_2[n] * h_2[n]$, $h_2[n] = u[n] - u[n-2]$, and $h[n] = \{1, 5, 10, 11, 8, 4, 1\}$ for $n = 0, \ldots, 6$

Step 1: Define $h_2[n]$

$h_2[n]$ is a rectangular pulse of length 2:

$$h_2[n] = \begin{cases} 1, & n = 0, 1 \\ 0, & \text{otherwise} \end{cases}$$

$h_2[n]$

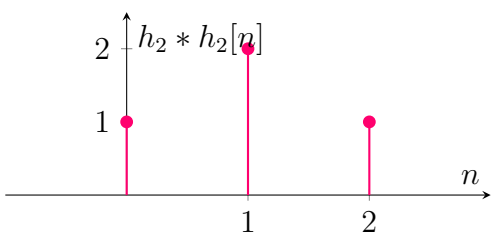


Step 2: Convolve $h_2[n] * h_2[n]$

Convolution of two length-2 rectangular pulses gives a triangular pulse of length 3:

$$(h_2 * h_2)[n] = \begin{cases} 1, & n = 0 \\ 2, & n = 1 \\ 1, & n = 2 \\ 0, & \text{otherwise} \end{cases}$$

$h_2[n] * h_2[n]$



Step 3: Determine $h_1[n]$ by Deconvolution

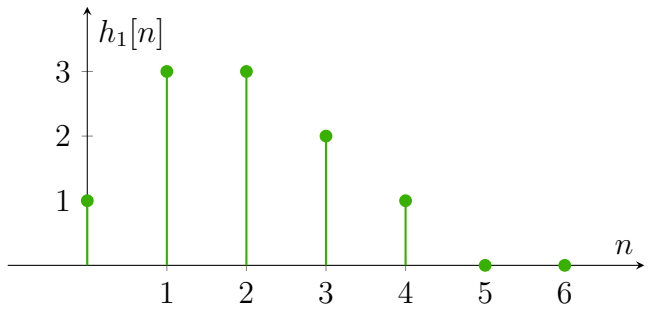
We have $h[n] = h_1[n] * (h_2 * h_2)[n]$ with $(h_2 * h_2)[n] = \{1, 2, 1\}$.

Solving iteratively:

$$
\begin{aligned}
n=0: &\quad h[0]=h_1[0]\cdot 1=1 \implies h_1[0]=1\\
n=1: &\quad h[1]=h_1[0]\cdot 2+h_1[1]\cdot 1=5 \implies 2+h_1[1]=5 \implies h_1[1]=3\\
n=2: &\quad h[2]=h_1[0]\cdot 1+h_1[1]\cdot 2+h_1[2]\cdot 1=10 \implies 1+6+h_1[2]=10 \implies h_1[2]=3\\
n=3: &\quad h[3]=h_1[1]\cdot 1+h_1[2]\cdot 2+h_1[3]\cdot 1=11 \implies 3+6+h_1[3]=11 \implies h_1[3]=2\\
n=4: &\quad h[4]=h_1[2]\cdot 1+h_1[3]\cdot 2+h_1[4]\cdot 1=8 \implies 3+4+h_1[4]=8 \implies h_1[4]=1\\
n=5: &\quad h[5]=h_1[3]\cdot 1+h_1[4]\cdot 2+h_1[5]\cdot 1=4 \implies 2+2+h_1[5]=4 \implies h_1[5]=0\\
n=6: &\quad h[6]=h_1[4]\cdot 1+h_1[5]\cdot 2+h_1[6]\cdot 1=1 \implies 1+0+h_1[6]=1 \implies h_1[6]=0
\end{aligned}
$$

$$\boxed{h_1[n]=\{1,3,3,2,1,0,0\}}$$

$h_1[n]$ (Deconvolution Result)



Step 4: Output for $x[n]=\delta[n]-\delta[n-1]$

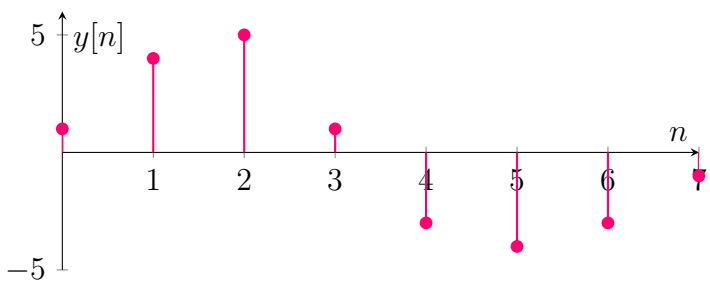
The output is:

$$y[n]=x[n]*h[n]=h[n]-h[n-1]$$

$$
\begin{aligned}
y[0]&=h[0]-0=1\\
y[1]&=h[1]-h[0]=5-1=4\\
y[2]&=h[2]-h[1]=10-5=5\\
y[3]&=h[3]-h[2]=11-10=1\\
y[4]&=h[4]-h[3]=8-11=-3\\
y[5]&=h[5]-h[4]=4-8=-4\\
y[6]&=h[6]-h[5]=1-4=-3\\
y[7]&=0-h[6]=-1
\end{aligned}
$$

$$\boxed{y[n]=\{1,4,5,1,-3,-4,-3,-1\}}$$

Output $y[n]=h[n]-h[n-1]$



Exercise 7: Signal Transformations

(a) Signal Transformation: $y[n] = 2r[1-n]$

The unit ramp is defined as $r[n] = n$ for $n \geq 0$, and 0 otherwise.

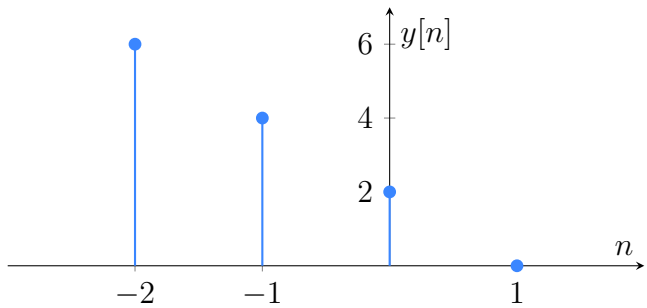
$$y[n] = 2r[1-n]$$

Step 1: $r[1-n] = 1-n$ for $1-n \geq 0$
Step 2: $1-n \geq 0 \implies n \leq 1$
Step 3: Apply scaling factor 2:

$$y[n] = \begin{cases} 2(1-n), & n \leq 1 \\ 0, & n > 1 \end{cases}$$

Values: $y[1] = 0$, $y[0] = 2$, $y[-1] = 4$, $y[-2] = 6$

$y[n] = 2r[1-n]$



(b) Expressing Signals using $\delta[n]$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

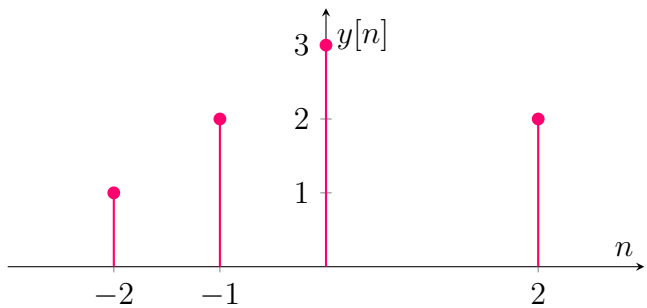
First signal: Non-zero samples: $n = 0$ (1), $n = 3$ (2).

$$x[n] = 1\delta[n] + 2\delta[n-3]$$

Second signal: Non-zero samples: $n = -2$ (1), $n = -1$ (2), $n = 0$ (3), $n = 2$ (2).

$$y[n] = 1\delta[n+2] + 2\delta[n+1] + 3\delta[n] + 2\delta[n-2]$$

Second Signal in $\delta[n]$ Form



(c) Sketching Signal Transformations

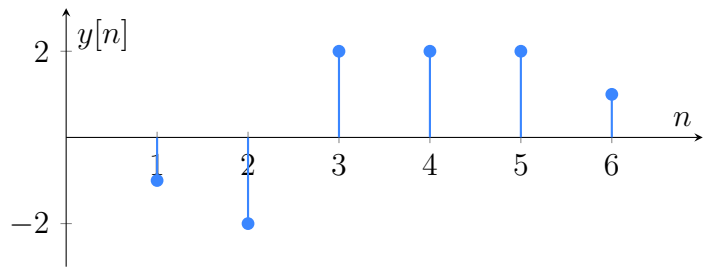
Given: $x[-2] = -1$, $x[-1] = -2$, $x[0] = 2$, $x[1] = 2$, $x[2] = 2$, $x[3] = 1$, all others 0

1. $y[n] = x[n-3]$ **(Time Shifting)**

Right shift by 3 units:

$$y[1] = -1,\ y[2] = -2,\ y[3] = 2,\ y[4] = 2,\ y[5] = 2,\ y[6] = 1$$

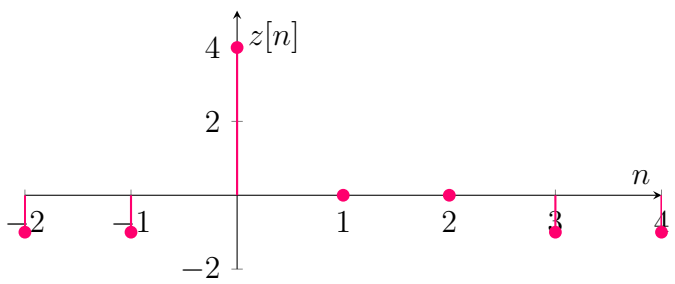
$y[n] = x[n-3]$ (Time Shift)



2. $z[n] = x[n] - x[n-1]$ **(First Difference)**

$$\begin{aligned}
z[-2] &= x[-2] - x[-3] = -1 - 0 = -1 \\
z[-1] &= x[-1] - x[-2] = -2 - (-1) = -1 \\
z[0] &= x[0] - x[-1] = 2 - (-2) = 4 \\
z[1] &= x[1] - x[0] = 2 - 2 = 0 \\
z[2] &= x[2] - x[1] = 2 - 2 = 0 \\
z[3] &= x[3] - x[2] = 1 - 2 = -1 \\
z[4] &= x[4] - x[3] = 0 - 1 = -1
\end{aligned}$$

$z[n] = x[n] - x[n-1]$ (First Difference)



Exercise 8: Linearity Determination

A system $\mathcal{T}$ is **linear** if it satisfies both:

1. **Additivity:** $\mathcal{T}(x_1[n] + x_2[n]) = \mathcal{T}(x_1[n]) + \mathcal{T}(x_2[n])$

2. **Homogeneity:** $\mathcal{T}(\alpha x[n]) = \alpha \mathcal{T}(x[n])$

System 1: $\mathcal{T}_1(x[n]) = x[n]u[n]$

$$\mathcal{T}_1(x_1 + x_2) = (x_1 + x_2)u[n] = x_1 u[n] + x_2 u[n] = \mathcal{T}_1(x_1) + \mathcal{T}_1(x_2)$$
$$\mathcal{T}_1(\alpha x) = (\alpha x)u[n] = \alpha x u[n] = \alpha \mathcal{T}_1(x)$$

$$\boxed{\textbf{Linear}}$$

System 2: $\mathcal{T}_2(x[n]) = x[n] + nx[n+1]$

$$\mathcal{T}_2(x_1 + x_2) = (x_1 + x_2) + n(x_1[n+1] + x_2[n+1]) = \mathcal{T}_2(x_1) + \mathcal{T}_2(x_2)$$
$$\mathcal{T}_2(\alpha x) = \alpha x + n\alpha x[n+1] = \alpha \mathcal{T}_2(x)$$

$$\boxed{\textbf{Linear}}$$

System 3: $\mathcal{T}_3(x[n]) = 2x[n-2] + 5$

$$\mathcal{T}_3(\alpha x) = 2\alpha x[n-2] + 5 \neq \alpha(2x[n-2] + 5) = \alpha \mathcal{T}_3(x)$$

$$\boxed{\textbf{Nonlinear (constant term breaks homogeneity)}}$$

System 4: $\mathcal{T}_4(x[n]) = \sum_{k=-5}^{\infty} 2x[k]$

$$\mathcal{T}_4(x_1 + x_2) = \sum_{k=-5}^{\infty} 2(x_1[k] + x_2[k]) = \sum 2x_1[k] + \sum 2x_2[k] = \mathcal{T}_4(x_1) + \mathcal{T}_4(x_2)$$
$$\mathcal{T}_4(\alpha x) = \sum_{k=-5}^{\infty} 2(\alpha x[k]) = \alpha \sum 2x[k] = \alpha \mathcal{T}_4(x)$$

$$\boxed{\textbf{Linear}}$$

System 5: $\mathcal{T}_5(x[n]) = 3x^2[n]$

$$\mathcal{T}_5(\alpha x) = 3(\alpha x)^2 = 3\alpha^2 x^2 \neq \alpha \mathcal{T}_5(x)$$

$$\boxed{\textbf{Nonlinear (squaring breaks homogeneity)}}$$

Exercise 9: Time-Invariance

A system $\mathcal{L}$ is **time-invariant** if for any shift n_0:

$$x[n - n_0] \xrightarrow{\mathcal{L}} y[n - n_0]$$

(a) $y[n] = \mathcal{L}(x[n]) = 10\sin(0.1\pi n)x[n]$

Shift input: $x_1[n] = x[n - n_0]$

$$y_1[n] = \mathcal{L}(x_1[n]) = 10\sin(0.1\pi n)x[n - n_0]$$

Shift output: $y[n - n_0] = 10\sin(0.1\pi(n - n_0))x[n - n_0]$
Since $10\sin(0.1\pi n)x[n - n_0] \neq 10\sin(0.1\pi(n - n_0))x[n - n_0]$

$$\boxed{\textbf{Time-Variant}}$$

(b) $y[n] = \mathcal{L}(x[n]) = x[n + 1] - x[1 - n]$

Shift input: $x_1[n] = x[n - n_0]$

$$y_1[n] = x[n - n_0 + 1] - x[1 - (n - n_0)] = x[n - n_0 + 1] - x[1 - n + n_0]$$

Shift output: $y[n - n_0] = x[n - n_0 + 1] - x[1 - (n - n_0)] = x[n - n_0 + 1] - x[1 - n + n_0]$
Since $y_1[n] = y[n - n_0]$

$$\boxed{\textbf{Time-Invariant}}$$

(c) $y[n] = \mathcal{L}(x[n]) = 14x[n] + 12x[n - 1] + 14x[n - 2]$

Shift input: $x_1[n] = x[n - n_0]$

$$y_1[n] = 14x[n - n_0] + 12x[n - n_0 - 1] + 14x[n - n_0 - 2]$$

Shift output: $y[n - n_0] = 14x[n - n_0] + 12x[n - n_0 - 1] + 14x[n - n_0 - 2]$
Since $y_1[n] = y[n - n_0]$

$$\boxed{\textbf{Time-Invariant}}$$

Exercise 10: System Coefficients

Given: $y[n] = 0.25y[n - 1] + x[n]$

Standard Form

General linear constant-coefficient difference equation:

$$a_0y[n] + a_1y[n - 1] + \cdots + a_Ny[n - N] = b_0x[n] + b_1x[n - 1] + \cdots + b_Mx[n - M]$$

Rewrite Given Equation

$$y[n] - 0.25y[n-1] = x[n]$$

Comparing with general form:

$$\begin{aligned} a_0 &= 1 \\ a_1 &= -0.25 \\ b_0 &= 1 \end{aligned}$$

All other coefficients are zero:

$$a_2 = a_3 = \cdots = 0, \quad b_1 = b_2 = \cdots = 0$$

$$\boxed{N = 1,\ M = 0,\ a_0 = 1,\ a_1 = -0.25,\ b_0 = 1}$$

Exercise 11: LTI System Analysis

Given: $y[n] = 0.5x[n] + 0.25x[n-1]$, $x[-1] = 0$

(a) Unit-Impulse Response $h[n]$

The impulse response is the output when $x[n] = \delta[n]$.

$$x[n] = \delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases}$$

For $n = 0$:

$$h[0] = 0.5 \cdot \delta[0] + 0.25 \cdot \delta[-1] = 0.5 \cdot 1 + 0.25 \cdot 0 = 0.5$$

For $n = 1$:

$$h[1] = 0.5 \cdot \delta[1] + 0.25 \cdot \delta[0] = 0.5 \cdot 0 + 0.25 \cdot 1 = 0.25$$

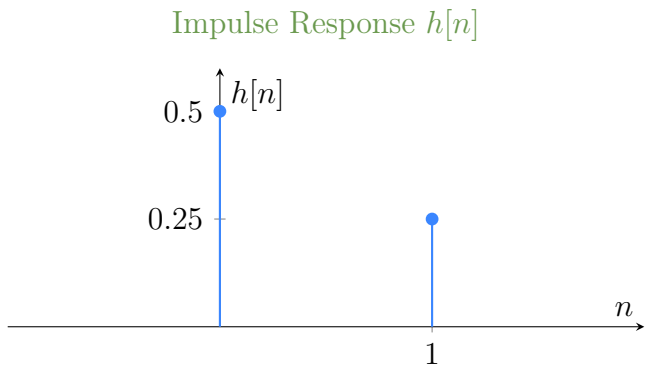
For $n \neq 0, 1$:

$$h[n] = 0.5 \cdot \delta[n] + 0.25 \cdot \delta[n-1] = 0 + 0 = 0$$

Therefore:

$$\boxed{h[n] = \begin{cases} 0.5, & n = 0 \\ 0.25, & n = 1 \\ 0, & \text{otherwise} \end{cases}}$$

In compact form: $h[n] = 0.5\delta[n] + 0.25\delta[n-1]$



(b) System Block Diagram

The system is described by: $h[n] = 0.5\delta[n] + 0.25\delta[n-1]$

$$x[n] \rightarrow \boxed{h[n] = 0.5\delta[n] + 0.25\delta[n-1]} \rightarrow y[n]$$

(c) Output Using Impulse Response

For an LTI system, the output is the convolution of the input with the impulse response:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

Step 1: Substitute $h[n] = 0.5\delta[n] + 0.25\delta[n-1]$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,(0.5\delta[n-k] + 0.25\delta[n-k-1])$$

Step 2: Use the distributive property to split the sum:

$$y[n] = 0.5 \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] + 0.25 \sum_{k=-\infty}^{\infty} x[k]\delta[n-k-1]$$

Step 3: Apply the sifting property of the delta function:

$$\sum_{k=-\infty}^{\infty} x[k]\delta[n-k] = x[n]$$

$$\sum_{k=-\infty}^{\infty} x[k]\delta[n-k-1] = x[n-1]$$

Step 4: Substitute back:

$$y[n] = 0.5 \cdot x[n] + 0.25 \cdot x[n-1]$$

This exactly matches the original difference equation, confirming our result.

$$\boxed{y[n] = 0.5x[n] + 0.25x[n-1]}$$

Exercise 12: Digital Convolution

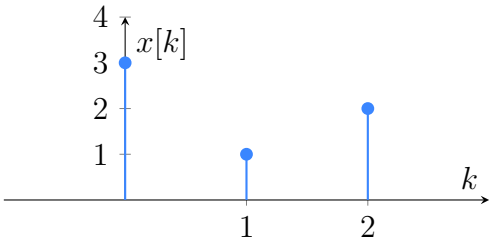
Given: $h[k] = \{3, 2, 1\}$ for $k = 0, 1, 2$, $x[k] = \{3, 1, 2\}$ for $k = 0, 1, 2$

Convolution length: $L_y = L_x + L_h - 1 = 3 + 3 - 1 = 5$, $n = 0, 1, 2, 3, 4$

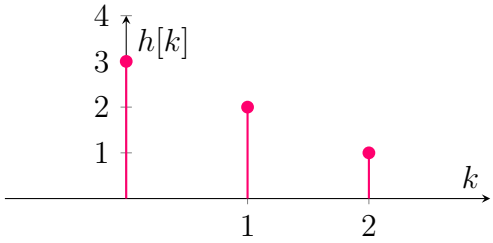
(a) Graphical Method (Flip and Slide)

$$
\begin{aligned}
y[0] &= x[0]h[0] = 3 \cdot 3 = 9 \\
y[1] &= x[0]h[1] + x[1]h[0] = 3 \cdot 2 + 1 \cdot 3 = 9 \\
y[2] &= x[0]h[2] + x[1]h[1] + x[2]h[0] = 3 \cdot 1 + 1 \cdot 2 + 2 \cdot 3 = 11 \\
y[3] &= x[1]h[2] + x[2]h[1] = 1 \cdot 1 + 2 \cdot 2 = 5 \\
y[4] &= x[2]h[2] = 2 \cdot 1 = 2
\end{aligned}
$$

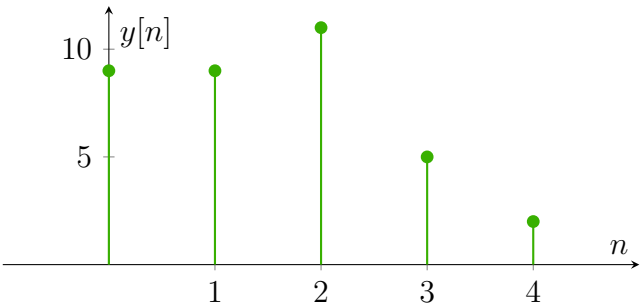
$$\boxed{y[n] = \{9, 9, 11, 5, 2\}, \quad n = 0, 1, 2, 3, 4}$$



Input Signal $x[k]$



Impulse Response $h[k]$



Convolution Result $y[n]$

(b) Direct Formula Method

Using tabular method (mirror matrix):

$x[k] \setminus h[k]$	3	2	1
3	9	6	3
1	3	2	1
2	6	4	2

Sum along anti-diagonals:

$$
\begin{aligned}
y[0] &= 9 \\
y[1] &= 6 + 3 = 9 \\
y[2] &= 3 + 2 + 6 = 11 \\
y[3] &= 4 + 1 = 5 \\
y[4] &= 2
\end{aligned}
$$

$$\boxed{y[n] = \{9, 9, 11, 5, 2\}}$$

Exercise 13: System Stability

Given: $y[n] = 0.5x[n] + 100x[n-2] + 20x[n-10]$

Stability Condition

A discrete-time LTI system is stable if its impulse response is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Impulse Response

For $y[n] = \sum_k b_k x[n-k]$, the impulse response is:

$$h[n] = 0.5\delta[n] + 100\delta[n-2] + 20\delta[n-10]$$

Absolute Sum

$$\sum_{n=-\infty}^{\infty} |h[n]| = |0.5| + |100| + |20| = 0.5 + 100 + 20 = 120.5 < \infty$$

The system is stable.

Exercise 14: Signal Classification

Classification of Signals

A signal can be classified into **one of three categories** based on its energy and power:

Energy Signal	Power Signal	Neither
• $0 < E < \infty$ • $P = 0$	• $0 < P < \infty$ • $E = \infty$	• $E = \infty$ • P undefined or ∞

Energy and Power Formulas

Continuous-Time

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$P = \lim_{T\to\infty} \frac{1}{T} \int_0^T |x(t)|^2 dt$$

Discrete-Time

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

Note: A signal belongs to exactly one category - it cannot be both an energy signal and a power signal.

Signal Classification

1. $x(t) = e^{-at},\ t > 0,\ a > 0$

$$E = \int_0^{\infty} e^{-2at} dt = \left[-\frac{e^{-2at}}{2a}\right]_0^{\infty} = \frac{1}{2a} - 0 = \frac{1}{2a} < \infty$$

$$P = \lim_{T\to\infty} \frac{1}{T} \int_0^T e^{-2at} dt = \lim_{T\to\infty} \frac{1}{T} \cdot \frac{1 - e^{-2aT}}{2a} = 0$$

$$\boxed{\textbf{Energy signal } (E = 1/2a,\ P = 0)}$$

2. $x(t) = A\cos(\omega t + \theta)$

$$E = \int_{-\infty}^{\infty} A^2 \cos^2(\omega t + \theta) dt = \infty \quad \text{(periodic, infinite duration)}$$

$$P = \lim_{T\to\infty} \frac{1}{T} \int_0^T A^2 \cos^2(\omega t + \theta) dt = \frac{A^2}{2}$$

$$\boxed{\textbf{Power signal } (E = \infty,\ P = A^2/2)}$$

3. $x[n] = 10e^{j2n}$

$$E = \sum_{n=-\infty}^{\infty} |10e^{j2n}|^2 = \sum_{n=-\infty}^{\infty} 100 = \infty$$

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} 100 = \lim_{N\to\infty} \frac{100(2N+1)}{2N+1} = 100$$

$$\boxed{\textbf{Power signal } (E = \infty,\ P = 100)}$$

4. $x(t) = t,\ t > 0$

$$E = \int_0^{\infty} t^2 dt = \lim_{b\to\infty} \left[\frac{t^3}{3}\right]_0^b = \infty$$

$$P = \lim_{T\to\infty} \frac{1}{T} \int_0^T t^2 dt = \lim_{T\to\infty} \frac{1}{T} \cdot \frac{T^3}{3} = \lim_{T\to\infty} \frac{T^2}{3} = \infty$$

$$\boxed{\textbf{Neither } (E = \infty,\ P = \infty)}$$

5. $x(t) = e^{-3t}u(t)$

$$E = \int_0^\infty e^{-6t}dt = \left[-\frac{e^{-6t}}{6}\right]_0^\infty = \frac{1}{6} < \infty$$

$$P = \lim_{T\to\infty} \frac{1}{T}\int_0^T e^{-6t}dt = \lim_{T\to\infty} \frac{1}{T} \cdot \frac{1 - e^{-6T}}{6} = 0$$

$$\boxed{\textbf{Energy signal } (E = 1/6,\ P = 0)}$$

6. $x(t) = (e^{-2t} + 1)u(t)$

$$E = \int_0^\infty (e^{-2t} + 1)^2 dt = \int_0^\infty (e^{-4t} + 2e^{-2t} + 1)dt$$

$$= \left[-\frac{e^{-4t}}{4} - e^{-2t} + t\right]_0^\infty = \infty \quad (\text{term } t \to \infty)$$

$$P = \lim_{T\to\infty} \frac{1}{T}\int_0^T (e^{-4t} + 2e^{-2t} + 1)dt$$

$$= \lim_{T\to\infty} \frac{1}{T}\left[-\frac{e^{-4t}}{4} - e^{-2t} + t\right]_0^T$$

$$= \lim_{T\to\infty} \frac{1}{T}\left(T - \frac{1}{4} - 1 + \frac{e^{-4T}}{4} + e^{-2T}\right)$$

$$= \lim_{T\to\infty} \left(1 - \frac{5}{4T} + \frac{e^{-4T}}{4T} + \frac{e^{-2T}}{T}\right) = 1$$

$$\boxed{\textbf{Power signal } (E = \infty,\ P = 1)}$$

7. $x(t) = t^{-1/2}u(t-2)$

$$E = \int_2^\infty |t^{-1/2}|^2 dt = \int_2^\infty t^{-1}dt = \lim_{b\to\infty} [\ln t]_2^b = \infty$$

$$P = \lim_{T\to\infty} \frac{1}{T}\int_2^T t^{-1}dt = \lim_{T\to\infty} \frac{\ln T - \ln 2}{T} = 0$$

$$\boxed{\textbf{Neither } (E = \infty,\ P = 0)}$$